

Musa Abdulrafiu

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Education

Fisk University

Nashville, TN

Bachelor of Science in Computer Science

Expected Graduation: May 2027

- Relevant Coursework: Data Structures and Algorithms, Computer Organization, Calculus I & II
- Honors/Awards: Erastus Milo Cravath Presidential Scholarship (\$138,000)

Technical Skills

Programming Languages: Python, Java, C++, JavaScript, TypeScript, Kotlin

Web & Mobile Development: React, React Native, Angular, Node.js, HTML, CSS

Backend & Databases: Express.js, PostgreSQL, MongoDB, Firebase (Auth, Firestore, Cloud Functions), Redis

Cloud & Infrastructure: AWS (EC2, S3, Lambda, API Gateway, CloudWatch), Docker, CI/CD, Linux/Unix

Software Engineering: OOP/OOD, Design Patterns, Distributed Systems, SDLC, Microservices

Tools & Technologies: Git, REST APIs, Jest, Jasmine, Karma

Experience

Google Inc.

San Francisco, CA

Associate Software Developer Intern

May 2025 – August 2025

- Built and launched the Experiment Hover Card for Primes UI using Angular and TypeScript, reducing experiment discovery time by 40%.
- Developed the feature from skeleton component to full functionality, including rollout support, error handling, and loading/empty states, reducing UI failure occurrences by 30%.
- Used metrics and logs to guide feature decisions and validate performance improvements.
- Wrote 100+ unit and integration tests with Jasmine/Karma, increasing component coverage from 78% to 96%.

Fisk University Department of Computer Science

Nashville, TN

AI/ML and Cyber-Physical Power Systems Security Research Intern

June 2024 – August 2024

- Applied supervised learning techniques, including CNNs, to analyze 27 videos and 40,000 frames from U.S. Department of Energy data for defect detection in pressure vessels and superheaters.
- Built a deep learning pipeline in Python using TensorFlow InceptionV3 for transfer learning and OpenCV for automated image processing.
- Integrated Stochastic Gradient Descent (SGD), achieving 99.47% accuracy even in challenging detection cases.

Projects

FiskConnect - Full-Stack Mobile Application

React Native, Firebase, Firestore, Expo, React Navigation, Git

- Built a campus-only mobile application for Fisk University students, restricting access to verified @fisk.edu accounts to create a trusted peer-to-peer marketplace and task coordination platform.
- Integrated Firebase Authentication, Cloud Firestore, and real-time onSnapshot listeners to deliver live updates for listings, tasks, and messages in 1–2 seconds without manual refresh.
- Designed secure Firestore data models and access rules for users, listings, tasks, conversations, and nested message threads; validated functionality across both Android and iOS devices.

ArtisanConnect - E-Marketplace for Handmade Goods

React, Node.js, Express.js, PostgreSQL, REST APIs, Git

- Architecting a scalable marketplace connecting U.S. buyers with global artisans selling handmade shoes, outfits, and custom products.
- Designed a PostgreSQL schema to support 1,000+ products, user accounts, order flows, and future platform growth.
- Built REST APIs with Node.js and Express for catalog management, user dashboards, and product interactions.
- Developed a responsive React front end for both desktop and mobile users, improving usability and accessibility.
- Leveraged Generative AI tools such as ChatGPT, Gemini, and Claude for test case generation, debugging, and faster iteration.

Finna Walk - Full-Stack Mobile App

React Native, Node.js, MongoDB, Firebase

- Built a full-stack fitness networking app enabling real-time student matching and session scheduling.
- Applied React Native concepts, including prop-based data transfer, useState, and useEffect, with collaborative development through Git.
- Applied OOP principles and modular component design across frontend and backend layers.
- Used Firebase Auth, Cloud Functions, Firestore, and security rules to authenticate users and securely manage student data.

Extracurriculars

- Fisk University Carl Van Vechten Gallery Ambassador.